23A31A0593

R23

Course Code:23BM304T

PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)

II B.Tech I Semester Regular Examinations, November-2024

DISCRETE MATHEMATICS AND GRAPH THEORY

(CSE, CSE(AI&ML), CSE(AI), CSE(DS), CSE(CS) and IT)

Time: 3 hours **Max. Marks: 70**

Note:

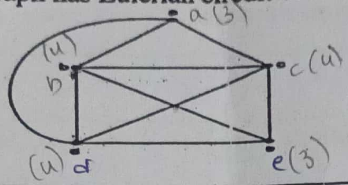
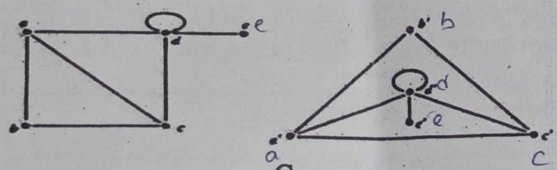
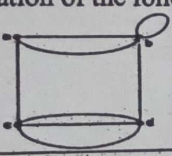
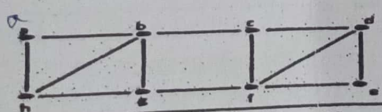
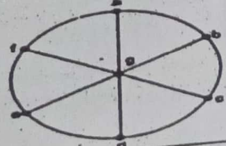
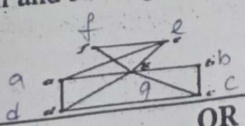
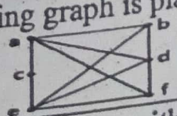
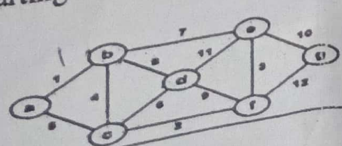
i. *Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.*

ii. *In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.*

Q. No.		Question	BTL	CO	Marks
1.	a)	Let the two statements be p: Mr. Bean learns discrete mathematics and q: Mr. Bean will find a good job. write the verbal sentence which describes (i) ~p and (ii) p → q.	K3	CO1	2M
	b)	Construct the truth table for the compound statement $q \leftrightarrow (\sim p \vee \sim q)$	K3	CO1	2M
	c)	Define compatibility relation.	K1	CO2	2M
	d)	If $f(x) = x^2 + 1$ and $g(x) = x+2$ are functions from R to R then find *fog and gof.*	K3	CO2	2M
	e)	How many ways can 5 programmers and 3 software engineers be seated around a circular table so that no two software engineers can sit together.	K2	CO3	2M
	f)	Find the number of possible arrangements of the letters in the word "DATABASES".	K2	CO3	2M
	g)	Define incidence matrix representation for undirected graph.	K1	CO4	2M
	h)	Define Hamiltonian graph with an example.	K1	CO4	2M
	i)	Find the Chromatic number of (i) cycle graph (Cn) and (ii) Complete bipartite graph ($K_{m,n}$)	K3	CO5	2M
	j)	Define planar graph with an example.	K1	CO5	2M
		UNIT-I			
2.	a)	Construct the truth table for the compound proposition $(p \wedge q) \rightarrow (p \vee q)$	K3	CO1	5M
	b)	Verify whether the premises "If you send me an e-mail message then I will finish writing the program. If you do not send me an e-mail message then I will go to sleep early. If I go to sleep early then I will wake up feeling refreshed." Lead to the conclusion "If I do not finish writing the program then I will wake up feeling refreshed."	K3	CO1	5M
		OR			
3.	a)	Prove that $p \vee (q \wedge r)$ is equivalent to $(p \vee q) \wedge (p \vee r)$ with the help of truth table.	K3	CO1	5M
	b)	Find the principal conjunctive normal form of the proposition $(p \wedge q) \vee (\sim p \vee q \vee r)$.	K3	CO1	5M
		UNIT-II			
4.	a)	Suppose that, a certain computer center employs 100 computer programmers. Of these 47 can program in Python, 35 in Java, 20 in Oracle. 23 can program in Python and Java, 12 in Python and Oracle, 11 in Java and Oracle and 5 in all the three. How many can program in none of these three?	K3	CO2	5M
	b)	Draw the Hasse diagram for the divisibility relation on the set A= {6, 12, 18, 24, 36, 72}.	K3	CO2	5M
		OR			
5.	a)	Consider the relation R = {(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)} defined on the set A = {1, 2, 3, 4}. (i) Find matrix representation of R (ii) Find R^{-1} and (iii) Draw the directed graph of R.	K3	CO2	6M
	b)	Let $f(x) = x + 3$, $g(x) = x - 4$ and $h(x) = 5x$ are functions from $R \rightarrow R$ where R is the set of real numbers. Find $fo(goh)$ and $(fog)oh$.	K3	CO2	4M

Course Code:23BM304T

R23

		UNIT-III			
6.	a)	How many arrangements are there of the letters "TALLAHASSEE" which have no adjacent A's.	K3	CO3	4M
	b)	Solve the recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 0$ for $n \geq 2$ with initial conditions $a_0 = 10$ and $a_1 = 41$ using generating function.	K3	CO3	6M
		OR			
7.	a)	Define multinomial theorem. Find the coefficient of $x\, y^2\, z^2\, w^3$ in the expansion of $(x + y + z + w)^8$.	K3	CO3	5M
	b)	Using the method of characteristic roots, solve the recurrence relation $a_n - 9a_{n-1} + 26a_{n-2} - 24a_{n-3} = 0,\ n \geq 3$.	K3	CO3	5M
		UNIT-IV			
8.	a)	Verify whether the following graph has Eulerian circuit or Eulerian path. 	K3	CO4	5M
	b)	Verify whether the following graphs are isomorphic or not.  G_1 G_2	K3	CO4	5M
		OR			
9.	a)	Write the adjacency matrix representation of the following graph. 	K3	CO4	5M
	b)	Verify whether the following graph has Hamiltonian cycle or not. If so, represent the Hamiltonian cycle. 	K3	CO4	5M
		UNIT-V			
10.	a)	Find the chromatic number of the following graph 	K3	CO5	5M
	b)	Using Depth first search algorithm and starting with vertex '*a*', find the spanning tree for the following graph. 	K3	CO5	5M
		OR			
11.	a)	Verify whether the following graph is planar or not. If so, draw the planar graph. 	K3	CO5	5M
	b)	Using Prim's algorithm and starting with vertex 'a', find the minimum spanning tree for the following graph. 	K3	CO5	5M